



**TARGET
BLUEYE**

PROJECT PLAN

Team – IT2A

“Track Technologies”

Team Members

Ai Nguyen (5473454)

Ana Olaru (5356768)

Arad Kashef Haghighi (5469236)

Derjen Frick (5588472)

Sara Kiani Nejad (5488621)

Yohan Lozanov (5213932)

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Version Control

This chapter keeps track of the document's version control.

Version	Date	Author(s)	Description
0.1	24.04.2026	Ai Nguyen, Ana Olaru, Arad Kashef Haghighi, Sara Kiani Nejad, Derjen Frick, Yohan Lozanov	Initial draft
0.2	07.05.2026	Yohan Lozanov	Added fix to Chapter 4.2. Now specifically requires app to device documentation. (based on client feedback)
0.3	08.05.2026	Arad Kashef Haghighi	Added paragraph to Chapter 2. Project goal

1. Background information

1.1. The Client: Track Technologies

Track Technologies is a private Dutch company specializing in the development of in-vehicle gadgets. Founded initially to produce accessories for motorcycles and other consumer automotive use cases, the company has focused exclusively on its flagship product, the Blue Eye, since 2013. Track Technologies develops all hardware and software in-house, with a compact development team responsible for both embedded systems and application software. Products are not sold directly to consumers but are instead distributed through third-party garage distributors who sell and install them in customer vehicles.

The primary contact person for this project is Frank Tieck, a software developer at Track Technologies who is responsible for both the Bluetooth communication backend and the coordination of the Android application project.

1.2 The Product: Blue Eye

The Blue Eye (Figure 1) is a dedicated in-vehicle device that passively detects radio signals transmitted on the C2000 network, the encrypted digital radio infrastructure used by Dutch emergency services including ambulances, police, and fire brigades. Because decoding the content of C2000 transmissions is prohibited by law, the Blue Eye does not decode these signals. Instead, it analyses signal strength, repetition frequency, and channel activity to estimate the likelihood and proximity of an approaching emergency vehicle.

When a signal is detected, the device provides a real-time alert to the driver using a combination of auditory and visual feedback. The current visual interface consists of a color-coded grid of LEDs (Figure 4) or an LCD panel (Figure 3), where indicators progress from green (initial detection, low confidence) through yellow to red (strong signal, high likelihood of an approaching vehicle). A maximum of three channels can be displayed simultaneously. The auditory alert is a repeating buzzer tone that activates when a signal is detected.

The Blue Eye exists in two hardware variants: a mobile dashboard unit designed to be mounted on the windscreen, and a built-in unit installed out of sight within the vehicle with only a separate screen visible to the driver. A Bluetooth communication module is currently in development and will enable wireless data exchange between the Blue Eye and external devices such as smartphones.

1.3 The Problem

Although the Blue Eye device functions reliably as a standalone unit, Track Technologies has identified a growing segment of customers, particularly drivers of premium vehicles, who prefer not to have a visible device mounted on their windscreen. These users would benefit from receiving the same real-time alert information directly on their smartphone, allowing the hardware unit to be concealed within the vehicle. In addition, the current physical interface offers limited configurability, adjusting device settings such as volume or display brightness require direct interaction with the hardware.

The absence of a companion Android application introduces several specific problems:

- **No mobile display:** There is no wireless interface for receiving real-time signal data from the Blue Eye device on a smartphone.
- **No remote configuration:** Users cannot adjust device settings such as volume, brightness, or alert preferences without physical access to the hardware.
- **Outdated interface:** The current visual display format (LED/LCD grid) is not easily transferable to a modern smartphone UI and has not been redesigned for mobile use.
- **Limited product extensibility:** The product cannot be marketed as a fully wireless, phone-based solution, limiting its appeal and future integration into newer vehicle form factors.

1.4 Expected Results and Success Criteria

The final product of this project will be a functional Android application that connects to the Blue Eye device via Bluetooth and serves as a real-time display and settings interface for the driver. The application will be delivered as a direct APK for testing purposes, with future publication to the Google Play Store handled by Track Technologies after handover.

Success will be measured against the following criteria:

- **Functional delivery:** The application establishes and maintains a Bluetooth connection to the Blue Eye device (or simulated equivalent via the provided Python test script) and displays incoming signal data in real time.
- **Core display:** Signal data including channel number and signal strength is clearly visualized on the main screen, in a manner that a driver can interpret instantly while keeping attention on the road.
- **Settings control:** The application allows the user to adjust at least the key device settings (e.g. volume, brightness) that are currently accessible on the hardware screen.
- **Connection state handling:** On launch, the application detects whether a Blue Eye device is connected and provides a clear status screen or warning before transitioning to the main view.

- **Design compliance:** The interface uses blue and black branding, prioritizes icons over text to support international use, and is designed for always-on display in a vehicle environment without causing distraction.
- **Timeline:** All in-scope features are implemented, tested against the Python simulation script, and delivered within the eight-week project period.

2. Project Objectives

2.1 Project goal

The overall goal of this project is to develop and deliver a functional Android companion application for the Blue Eye device within eight weeks, directly increasing Track Technologies' commercial reach by offering customers a wireless, phone-based alternative to the physical display unit. Success will be measured by the application's ability to establish a stable Bluetooth connection, display real-time C2000 signal data clearly on-screen, and allow basic device configuration, all verified through the provided Python simulation script and client acceptance testing. This is achievable within the project timeline given the existing Kotlin library and simulation tooling supplied by the client. The application addresses a demonstrated road safety need: drivers frequently fail to hear approaching emergency vehicles in time, and by making the Blue Eye's kilometer-range detection accessible without visible in-car hardware, the product becomes viable for a broader audience. Delivery of a tested APK along with the developer manual to Track Technologies by the end of Sprint 3 marks the completion of this goal.

2.2 Project Objectives

The main objective of this project is to design and develop a functional Android application that connects to the Blue Eye device via Bluetooth and presents real-time signal information to the user in a clear and usable way. The application should act as a digital extension of the existing hardware interface, while improving usability and modernizing the overall user experience.

To achieve this, the project focuses on the following objectives:

- **Develop a stable Bluetooth connection**
The application must be able to connect to the Blue Eye device (or its simulation) and reliably receive real-time signal data.
- **Ensure connection flow and device detection**
The application must include a clear connection process. On startup, a loading or scanning screen should inform the user that the app is searching for a Blue Eye device. Once a connection is successfully established, the user is automatically redirected to the main (home) screen. If no device is found, the app should clearly indicate this and allow retrying.
- **Create a clear real-time display**
Signal data such as channel activity and signal strength should be visualized in a simple, intuitive interface that can be quickly understood while driving.
- **Design a mobile-friendly user interface**
The interface should be optimized for in-vehicle use, using clear visuals, minimal text, and a layout that reduces distraction.

- **Provide basic device interaction (read-focused)**
The application should display device data and status clearly, staying within the defined scope of the project.
- **Align with branding and usability requirements**
The application should follow the blue/black design style and prioritize icons and visual cues to support international users.
- **Deliver a working prototype within the project timeline**
A complete APK must be delivered, tested using the provided simulation, and ready for handover within the eight-week period.

The overall objective is to transform the Blue Eye from a hardware-only product into a more flexible and modern solution by introducing a mobile companion that improves usability, accessibility, and future scalability.

3. Project Activities

This chapter will give a description of the activities needed to be done within the project to achieve the project goals. The work is divided into phases, following the Scrum technique.

3.1. Activity Overview

The project involves a combination of technical, integration and testing efforts, including:

Technical development:

- Bluetooth communication with the Target Blue Eye device
- Real-time signal display and always-on screen
- Sound alert integration
- Expandable language architecture

Testing:

- Feature testing using Python Bluetooth simulation script
- Internal integration testing per sprint
- User acceptance testing with Track Technologies

Communication and testing:

- Weekly client meetings with client (Track Technologies)
- Daily stand-ups within the team
- Sprint reviews and retrospectives

Documentation:

- Requirement analysis and project documentation
- Technical documentation of the app and Kotlin library integration
- Test documentation (test plan and test results)

Final delivery:

- APK build and delivery to Track Technologies
- Handover documentation for maintenance

This activity overview gives a clear and structured guideline for the team to track progress and realize what needs to be done at each stage to deliver a functional and customized time registration tool.

3.2. Work Breakdown Structure

The project tasks are divided into phases and grouped into individual activities in Table 1 to help the team understand the workflow, track progress, and assign responsibilities across the 8-week timeline.

Phase	Group Title	Activity
Sprint 0 Documentation (Week 1–2)	Planning	Requirement analysis and documentation
		Obtain Kotlin library documentation from client
		Obtain Python simulation script from client
	Setup	Set up development environment
		Study Kotlin library documentation
		Set up Bluetooth simulation (Python script)
Sprint 1 Core Development (Week 3–4)	Design	UI/UX design (blue/black, icon-based)
		App architecture design
	Bluetooth Communication	Implement Bluetooth communication layer
		Test Bluetooth with Python simulation
	Signal Display	Build real-time signal display screen
		Always-on display implementation
Sprint 2 Feature Development (Week 5–6)	UI & Features	Build settings/configuration screen
		Sound alert integration
		Expandable language architecture setup
	Testing	Sprint 2 internal testing

Phase	Group Title	Activity
Sprint 3 Testing & Delivery (Week 7–8)	Testing	Internal testing with Python simulation
		Bug fixes and refinements
		User acceptance testing with Track Technologies
	Delivery	Final documentation
		APK build and delivery to client

Table 1 - WBS

4. Project Boundaries

This chapter will describe the scope of the project and the conditions required for this project to succeed.

4.1. Project Scope

Activities within scope:

- Creation of the complete android app companion for the physical tracking device.
- Research regarding driver behavior with relevance to phone usage behind the wheel.
- Creation of language framework in app.
- Interfacing of app with the physical tracking device.
- Creation of analysis or implementations for the accessibility of the application for physically impaired user groups (like the deaf).

Activities outside of scope:

- Any and all alterations to the physical tracking device.
- Any and all alterations to the firmware or software of the physical tracking device.
- Giving suggestions or analysis about drivers not relating to the application.
- Giving suggestions or analysis regarding emergency vehicles.
- Implementation of concrete language support.
- Implementation of control over physical tracking device (app will be read-only).
- Creation of analysis or implementation for accessibility of the application for cognitively impaired user groups (including alcohol impairment).
- Live field testing. (Any tests performed which involve the use of an automobile will be performed in a controlled environment).
- The creation, alteration, or maintenance of any multi-user trackers.

4.2. Project conditions

For this project to succeed, the following parties must follow these conditions:

IT-2A:

- Complete Android application. Must be delivered on the 19th of June.
- Applicable documentation of the android application. Must be delivered on the 19th of June.
- Weekly demo. Must be planned by IT-2A for every week in the academic calendar within the duration of the project.

Track Technologies:

- Applicable available usage documentation for the connection between the app and physical tracking device. Must be delivered before the 11th of May.
- An available interface of the physical tracking device (for the purposes of testing). Must be made available before the 11th of May.

5. Organization

This chapter describes the overall organization of the team involved in the project, outlining their contact information, attributions and availability

5.1 The Team

Given the complexity of the project, its successful completion requires collaboration and teamwork. This approach enables efficient work, and a lower workload per team member, as the tasks can be distributed evenly. In order to effectively contribute to the project, each team member has been assigned a personal role, with specific attributions.

The general information about each team member, as well as their roles and contact details are listed in Table 2:

TEAM MEMBER NAME	ROLE	PHONE NUMBER	EMAIL
Frank Tieck	Client	+31 6	f.tieck@tracktechnologies.nl
Arad Kashef Haghighi	Product Owner	+31 6 33380280	flavius.petrasciuc@student.nhlstenden.com
Sara Kiani Nejad	Developer	+31 6 17375166	sara.kiani.nejad@student.nhlstenden.com
Ai Nguyen	Developer	+31 6 86213298	ai.nguyen@student.nhlstenden.com
Ana Olaru	Scrum Master	+40 769 193 124	ana.olaru@student.nhlstenden.com
Derjen Frick	Developer	+31 6 86133826	derjen.frick@student.nhlstenden.com
Yohan Lozanov	Developer	+31 6 87292302	yohan.lozanov@student.nhlstenden.com

Table 2 - Team member

5.2. Team Roles

PRODUCT OWNER

- Establishes contact between the team and the client, as well as other parties.
- Leads client meetings.

SCRUM MASTER

- Leads internal sprint planning and daily stand-up.
- Oversees the workflow within the team.
- Distribute tasks between team members.

DEVELOPER

- Designs and builds the product.
- Fixes reported bugs and errors reported by the quality controller.

5.3. Availability

The availability of the client, as well as each team member is illustrated in Table 3:

LEGEND	
	Available online
	Unavailable

NAME	MON	TUE	WED	THU	FRI	SAT	SUN
F. Tieck							
A. Kashef Haghighi							
S. Kiani Nejad							
A. Nguyen							
A. Olaru							
D. Frick							
Y. Lozanov							

Table 3 - Availability

6. Planning

This chapter outlines the planning strategy for the Target Blue Eye Android application project, providing a structured timeline that connects tasks to team members and milestones. It helps estimate the project's duration and ensures smooth coordination and accountability throughout the development process.

6.1. Activity Schedule Table

Table 4 summarizes the core activities to deliver the final product. Each task is assigned to a team member or team to help visualize the workflow.

Activity	Duration	Assigned to	Dependencies
Requirement analysis and documentation	2 weeks	Team + Client	–
Set up development environment	3 days	Team	Requirement analysis
Study Kotlin library documentation	1 week	Team	Library delivered by client
Set up Bluetooth simulation (Python)	2 days	Team	Python script delivered by client
UI/UX design	4 days	Team + Client	Requirement analysis
App architecture design	3 days	Team	Requirement analysis
Implement Bluetooth communication layer	1 week	Team	Kotlin library, Python simulation
Build real-time signal display screen	1 week	Team	Bluetooth layer, UI/UX design
Build settings/configuration screen	4 days	Team	Signal display screen
Always-on display implementation	3 days	Team	Signal display screen
Sound alert integration	2 days	Team	Signal display screen
Internal testing with Python simulation	3 days	Team	All development complete
Bug fixes and refinements	4 days	Team	Internal testing
User acceptance testing	3 days	Team + Client	Bug fixes complete
Final documentation	2 days	Team	User acceptance testing
APK build and delivery	1 day	Team + Client	Final documentation

Table 4 - Activity schedule

6.2. Milestones

Table 5 marks the significant achievements throughout the project. Each milestone represents the completion of a key phase or deliverable within the 8-week timeline.

Milestones	Description	Target date
M1: Project kick-off	Team assembled, Kotlin library and Python script received, requirements finalised, development environment ready	End of week 2
M2: Sprint 0 complete	All documentation finished, app architecture designed, UI/UX draft approved by client	End of week 2
M3: Bluetooth layer ready	Bluetooth communication with device simulation working, signal data successfully received by the app	End of week 3
M4: Core UI complete	Real-time signal display screen and always-on display implemented and functional	End of week 5
M5: Full app functional	Settings screen, sound alerts, and all core features working end-to-end with Python simulation	End of week 6
M6: Testing complete	All internal tests passed, user acceptance testing done with Track Technologies, bugs fixed	End of week 7
M7: Project delivery	Final documentation complete, APK built and delivered to Track Technologies	End of week 8

Table 5 - Milestones

6.3. Planning

Table 6 visualizes the duration of each phase as well as the achieved milestones and Table 7 demonstrates phase's span

Sprite/Phase	Milestone	How long it will take
Sprint 0 – Documentation	M1, M2	2 weeks
Sprint 1 – Core Development	M3, M4	2 weeks
Sprint 2 – Feature Development	M4, M5	2 weeks
Sprint 3 – Testing & Delivery	M6, M7	2 weeks

Table 6 - Planning

Main Milestone	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8
<i>Documentation</i>								
<i>Coret Development</i>								
<i>Feature Development</i>								
<i>Testing & Delivery</i>								

Table 7 - Gantt chart

7. References

7.1. List of devices

Blue Eye Go:



Figure 1 - Blue Eye Go. Source: <https://blu-eye.eu>

Blue Eye 2:



Figure 2 - Blue Eye 2. Source: <https://blu-eye.eu>

LCD screen:



Figure 3 - LCD screen. Source: <https://blu-eye.eu>

LED panel:

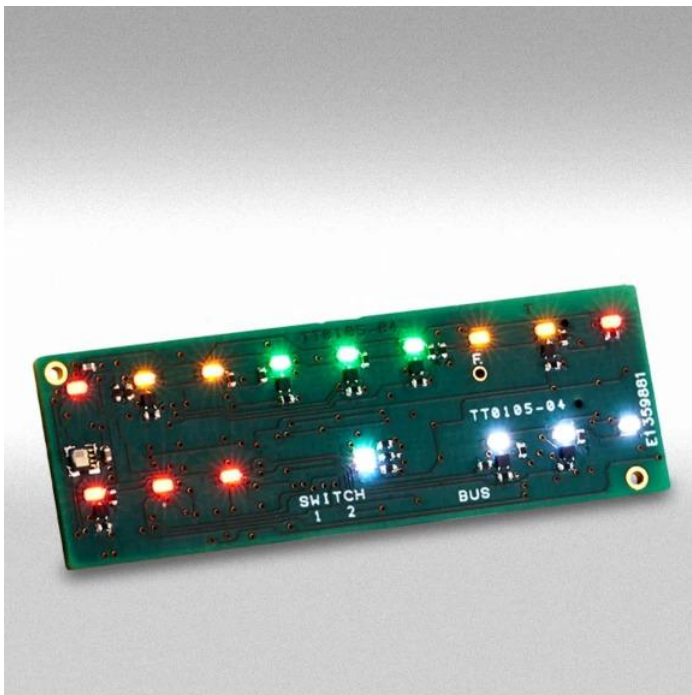


Figure 4 - LED. Source: <https://blu-eye.eu>

7.2. References

Blue Eye Go. Target Blue Eye. <https://blu-eye.eu/productinformatie/target-blu-eye-go>

Blue Eye 2. Target Blue Eye. <https://blu-eye.eu/en/product-information/target-blu-eye-2>

LCD screen. Target Blue Eye. <https://blu-eye.eu/en/shop/blu-eye-2-with-lcd-display>

LED. Target Blue Eye. <https://blu-eye.eu/en/shop/blu-eye-2-with-led-display>